

Demonstration of a Theorem of Abel

[Note by M. Lejeune Dirichlet communicated by M. Liouville.]

G. Lejeune Dirichlet, *Werke*, Band II, pp. 303–306.

The point is to prove that if the series

$$a_0 + a_1 + a_2 + \cdots + a_n + \cdots$$

is convergent and has sum A , then the sum of the series

$$a_0 + a_1\rho + a_2\rho^2 + \cdots + a_n\rho^n + \cdots,$$

which will be convergent *a fortiori* when the variable ρ is taken positive and less than unity, will tend to the limit A when ρ is made to tend indefinitely to unity.

One day, in conversation with my excellent and much-lamented friend Lejeune Dirichlet, I told him that I found it rather difficult to set out (and even to understand) the proof which Abel had given of this important theorem. Dirichlet at once began writing before my eyes, solely in order to help me, the following note, which has been of great assistance to me and which the public will be grateful to have. The method of proof found there admits numerous applications and has often been useful to me in my lectures at the Collège de France.

I transcribe Dirichlet's note textually, without adding anything to it and, of course, without changing anything.

“From the assumed convergence of the series

$$A = a_0 + a_1 + a_2 + \cdots + a_n + \text{etc.},$$

it follows that the sum

$$s_n = a_0 + a_1 + \cdots + a_n$$

always remains numerically below a certain constant k , and converges to the limit A when n increases indefinitely. Now consider the series

$$S = a_0 + a_1\rho + a_2\rho^2 + \cdots + a_n\rho^n + \text{etc.},$$

the quantity ρ being assumed positive and less than unity. Replacing a_0, a_1, a_2 , etc. in it by $s_0, s_1 - s_0, s_2 - s_1$, etc., it takes the form

$$S = s_0 + (s_1 - s_0)\rho + (s_2 - s_1)\rho^2 + \cdots + (s_n - s_{n-1})\rho^n + \text{etc.}$$

and then, by ordering otherwise,

$$S = (1 - \rho)(s_0 + s_1\rho + s_2\rho^2 + \cdots + s_n\rho^n + \cdots).$$

This transposition involves no difficulty, since it reduces to adding to the sum of the first $n + 1$ terms the term $-s_n\rho^{n+1}$, which vanishes for $n = \infty$.

“Let us now see to what limit S converges when the positive variable $\varepsilon = 1 - \rho$ becomes infinitely small. For this purpose decompose S into two parts, one comprising the first n terms and the other all the following terms, and let n grow as ε decreases, but slowly enough that the limit of $n\varepsilon$ is zero. The first part

$$(1 - \rho)(s_0 + s_1\rho + \cdots + s_{n-1}\rho^{n-1}),$$

being numerically less than $n\varepsilon k$, converges to zero. As for the second,

$$(1 - \rho)(s_n\rho^n + s_{n+1}\rho^{n+1} + \cdots),$$

one may give it the form

$$P(1 - \rho)(\rho^n + \rho^{n+1} + \dots) = P\rho^n = P(1 - \varepsilon)^n,$$

where P denotes a value lying between the greatest and the smallest of the quantities $s_n, s_{n+1}, \dots$. Now all these latter quantities converge to the limit A , and therefore so does P . On the other hand, the factor $(1 - \varepsilon)^n$, by virtue of the hypothesis made above, evidently converges to unity. Thus it is proved that the limit of S , when the positive variable $\rho < 1$ approaches unity indefinitely, is precisely the sum A of the series considered at the beginning."

I do not think that anyone can now think of asking for further clarification.